Icart-3 60W 0803

INTEGRATING YP-SPUR AND ROS2
(UBUNTU LINUX 22.04)

What was used

- I-Cart 3
- Ubuntu Linux 22.04
- ROS2 Humble
- YP-Spur(BLVR mode)
- Hokuyo URG (LiDAR)
- Logicool F710 controller

What needs to be installed

- Install ROS2 Humble from here
- If you are new to ROS2 there are useful tutorials on the page as well
- Next execute the following commands in the terminal:

```
1. sudo apt update
2. sudo apt install ros-humble-desktop
3. sudo apt install ros-humble-urg-node
4. sudo apt install ros-humble-cartographer
5. sudo apt install ros-humble-cartographer-ros
6. sudo apt install ros-humble-nav2-map-server
7. sudo apt install libmodbus-dev
8. sudo apt install ros-humble-teleop-twist-joy
9. sudo apt install python3-pip
10. pip3 install evdev
```

- Install the following YP-Spur (BLVR-compability needed for I-Cart 3): https://github.com/BND-tc/yp-spur

```
1. cd
2. git clone https://github.com/BND-tc/yp-spur.git
3. cd yp-spur
4. mkdir build
5. cd build
6. ../configure
7. make
8. sudo make install
```

What needs to be installed

- Install the YP-Spur – ROS2 bridge
- If you have not made a ROS2 workspace yet, do it now by executing the following commands:

```
1. mkdir -p ~/ros2_ws/src
2. cd ~/ros2_ws/src
```

- Then clone the repository and build:

```
1. git clone https://github.com/haruyama8940/icart_mini_driver_ros2.git
2. cd ~/ros2_ws
3. colcon build
4. source install/setup.bash
```

SLAM configuration my_robot_slam.lua

- Execute commands:

```
1. cd
2. nano my_robot_slam.lua
```

- Then copy the my_robot_slam lua code from here to that file

Cartographer configurations

- Execute commands:

```
1. cd
2. nano  cartographer_launch.py
```

- Then copy the configurations from here and paste it to cartographer_launch.py
- Note that you need to change the cartographer_config_dir according to your system

```
# 1. Configuration Variables
use_sim_time = False
cartographer_config_dir = '/home/niksal'
configuration_basename = 'my_robot_slam.lua'
resolution = '0.05'
publish_period_sec = '1.0'
```

Create a parameter file for the robot

- Execute commands:

```
1. cd
2. nano iCart3_60W0803.param
```

- Then copy the iCart3 60W0803 parameters from here and paste it to iCart3_60W0803.param

Starting the robot

- When starting for the first time execute command:

```
1. sudo ldconfig
```

- Terminal 1 (YP-Spur):

```
1. sudo ypspur-coordinator --blvr -p
~/iCart3_60W0803.param
```

- Terminal 2 (ROS2 driver):

```
1. source ~/ros2_ws/install/setup.bash
2. ros2 run icart_mini_driver icart_mini_driver
```

- Terminal 3 (Transforms / TF):

```
1. source /opt/ros/humble/setup.bash
2. ros2 run tf2_ros static_transform_publisher
 0.2 0 0 0 0 0 base_link laser & ros2 run tf2_ros
static_transform_publisher 0 0 0 0 0 0 base_footprint
base_link
```

- Terminal 4 (Controller):

```
1. source /opt/ros/humble/setup.bash
2. ros2 launch teleop_twist_joy teleop-launch.py
joy_config:='xbox'
```

- Terminal 5 (LiDAR):

```
1. sudo chmod 666 /dev/ttyACM*
2. source /opt/ros/humble/setup.bash
3. ros2 run urg_node urg_node_driver --ros-args -p
serial_port:=/dev/ttyACM* -p frame_id:=laser -p
angle_min:=-2.09 -p angle_max:=2.09
```

- Terminal 6 (Cartographer):

```
1. source /opt/ros/humble/setup.bash
2. ros2 launch ~/cartographer_launch.py
```

Starting the robot

- Terminal 7 (Visualization):
  Execute commands:

```
1. source /opt/ros/humble/setup.bash
2. rviz2
```

- Then add: laserscan, map, and odometry (optional), and set them to their respective topics (/scan, /map, /odom)
- Terminal 8 (Save map):

```
1. source /opt/ros/humble/setup.bash
2. ros2 run nav2_map_server map_saver_cli -f ~/my_map_name
```

- If the controls are inverted (Robot moves back when trying to go forward and moves left when trying to move right, etc...)
- Execute command:

```
1. nano ~/ros2_ws/src/icart_mini_driver_ros2/src/icart_mini_driver_node.cpp
```

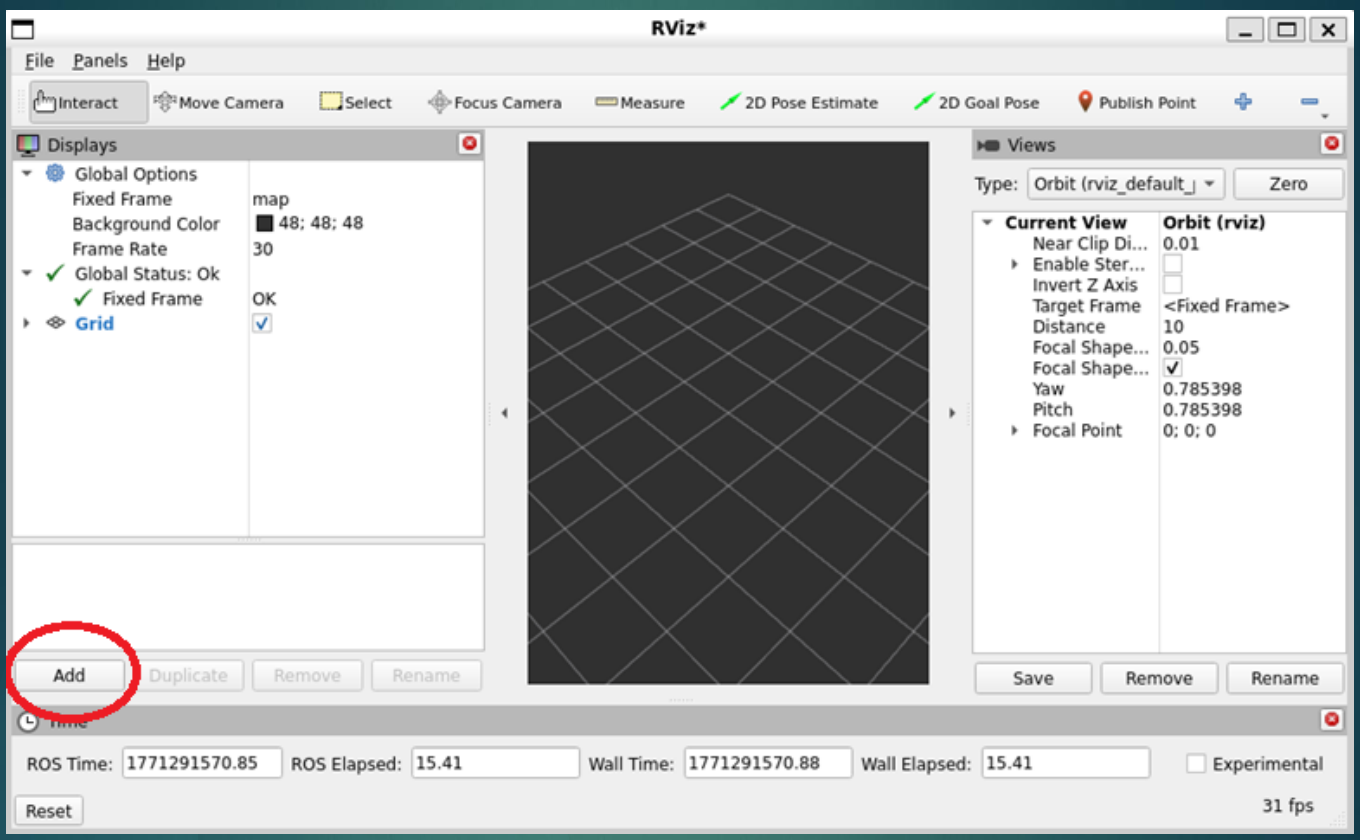
- Delete everything in the file above, then replace it with the icart_mini_driver_node.cpp code from here
- Then rebuild workspace:

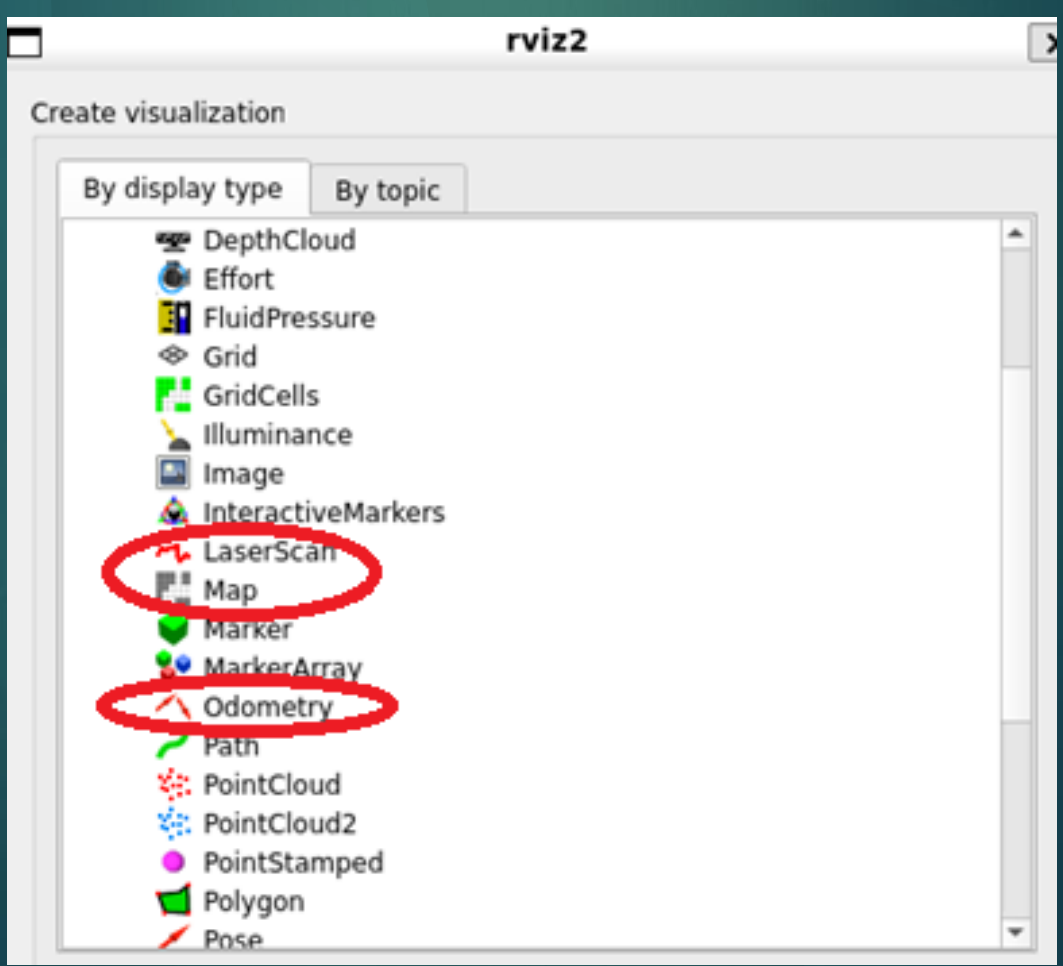
```
1. cd ~/ros2_ws
2. colcon build
```

- Now the controls should be correct!

RVIZ configurations for SLAM

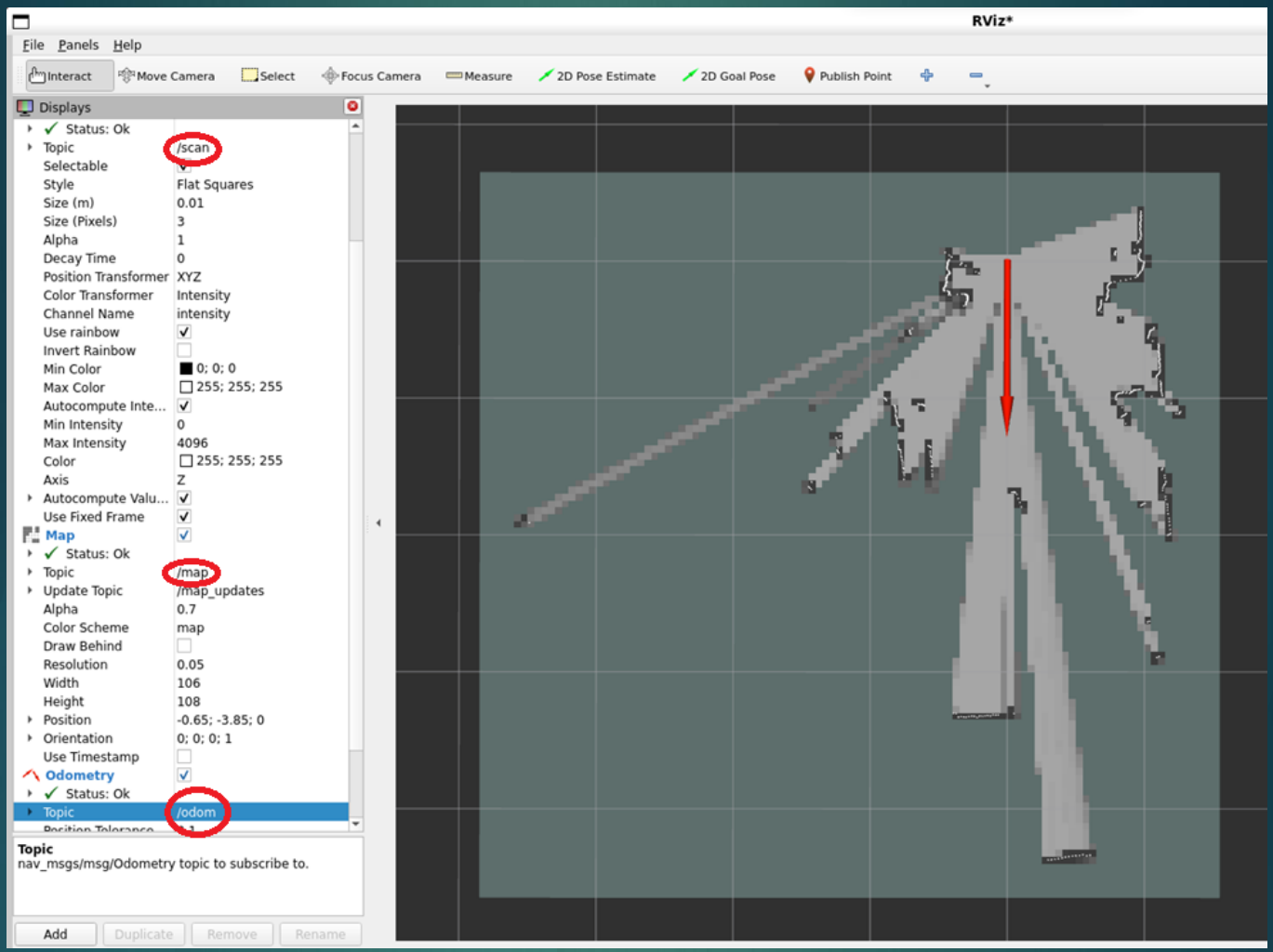
- First add LaserScan, Map and Odometry (Odometry is optional, but it shows the orientation of the robot on the map)





RVIZ configurations for SLAM

- Next select right topics for these
  LaserScan = /scan
  Map = /map
  Odometry = /odom



How to open many terminals simultaneously

- Custom script is a good option.
  Execute commands:

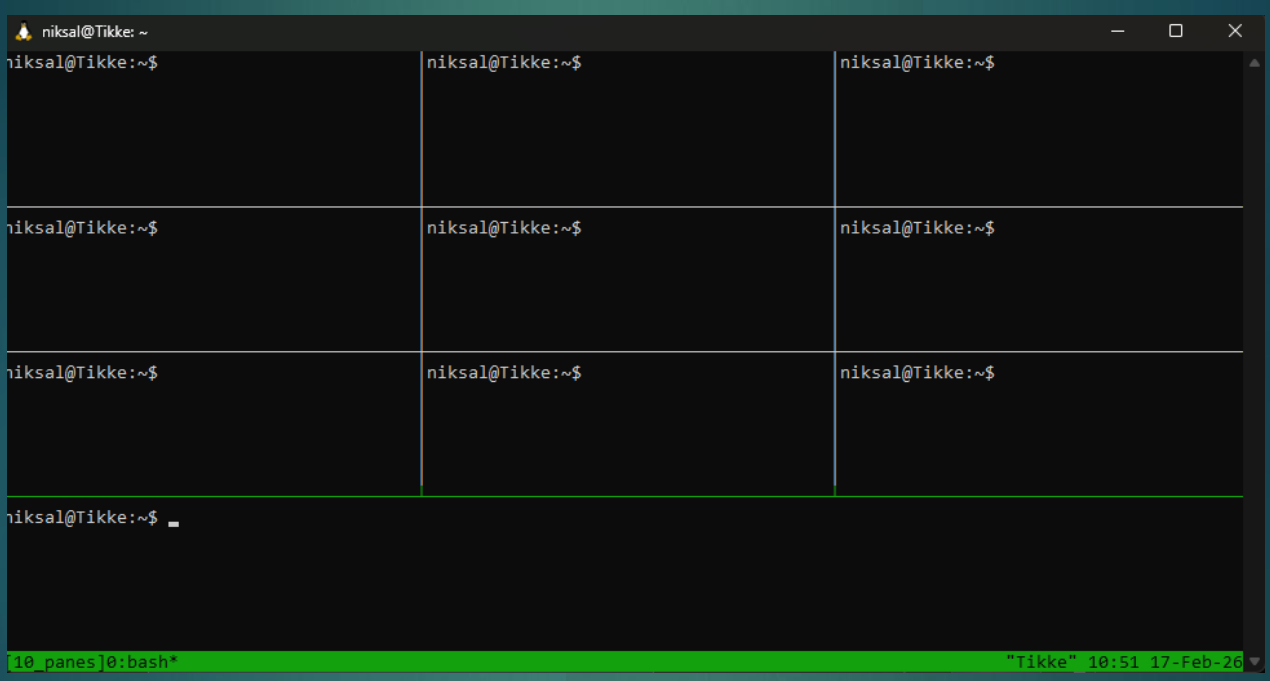
```
1. cd
2. nano open_10_panes.sh
```

- Copy the script from here to open_10_panes.sh
- Make the file executable by executing command:

```
1. chmod +x open_10_panes.sh
```

- You can now use the script by executing command:

```
1. ./open_10_panes.sh
```



- You can exit the 10 panes by executing command:

```
1. tmux kill-session -t 10_panes
```

Add a custom Navigation file

- The default navigation file is made for smaller, round robots like Turtlebot
- As Icart-3 is quite big, the parameters have to be modified
- I have prepared a pre-configured file with right size of the robot, and it is available here
- Use the following to create a config-directory to the icar_mini_driver folder, create the parameter file there and copy the code from the link above

1. mkdir -p ~/ros2_ws/src/icart_mini_driver_ros2/config
2. nano ~/ros2_ws/src/icart_mini_driver_ros2/config/icart_nav2_params.yaml

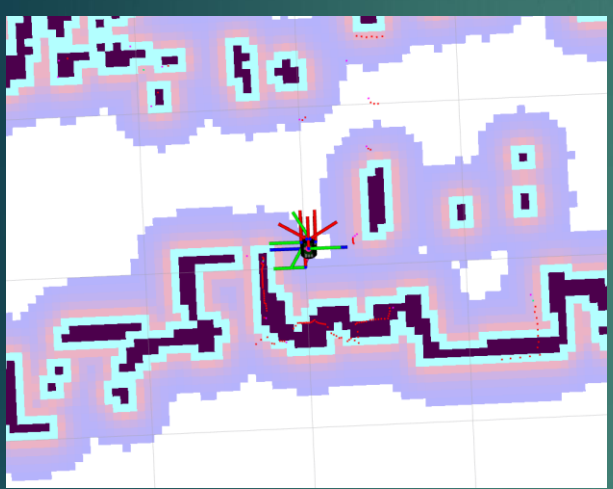
- Note that you might have to edit the file, max_vel_x, the cost scaling factors and inflation layer are very important
- Inflation layer is the size of the safety buffer around obstacles. If the robot hits objects, increase inflation_radius
- Cost scaling factor controls how far the robot tries to avoid obstacles. If the robot takes long paths, increase cost_scaling_factor

```
plugins: [ "voxel_layer", "inflation_layer" ]
inflation_layer:
  plugin: "nav2_costmap_2d::InflationLayer"
  cost_scaling_factor: 3.0
  inflation_radius: 0.55
voxel_layer:
```

Effect of inflation radius and cost scaling factor

- Low cost_scaling factor and high inflation_radius make moving hard for the robot in small spaces

```
inflation_layer:
  plugin: "nav2_costmap_2d::InflationLayer"
  cost_scaling_factor: 15.0
  inflation_radius: 0.30
```



- This can be improved with tuning the cost_scaling_factor up and inflation radius down

```
inflation_layer:
  plugin: "nav2_costmap_2d::InflationLayer"
  cost_scaling_factor: 25.0
  inflation_radius: 0.16
```

- The perfect values depend on the robot and the environment, and are only found with careful tuning

Navigation

- Terminal 1 (YP-Spur):

```
1. sudo ypspur-coordinator --blvr –p
~/iCart3_60W0803.param
```

- Terminal 2 (ROS2 driver):

```
1. source ~/ros2_ws/install/setup.bash
2. ros2 run icart_mini_driver icart_mini_driver
```

- Terminal 3 (LiDAR):

```
1. sudo chmod 666 /dev/ttyACM*
2. source /opt/ros/humble/setup.bash
3. ros2 run urg_node urg_node_driver --ros-args –p
serial_port:=/dev/ttyACM* -p frame_id:=laser –p
angle_min:=-2.09 -p angle_max:=2.09
```

- Terminal 4 (Transforms / TF):

```
1. source /opt/ros/humble/setup.bash
2. ros2 run tf2_ros static_transform_publisher
 0.2 0 0 0 0 0 base_link laser & ros2 run tf2_ros
static_transform_publisher 0 0 0 0 0 0 base_footprint
base_link
```

- Terminal 5 (Navigation):

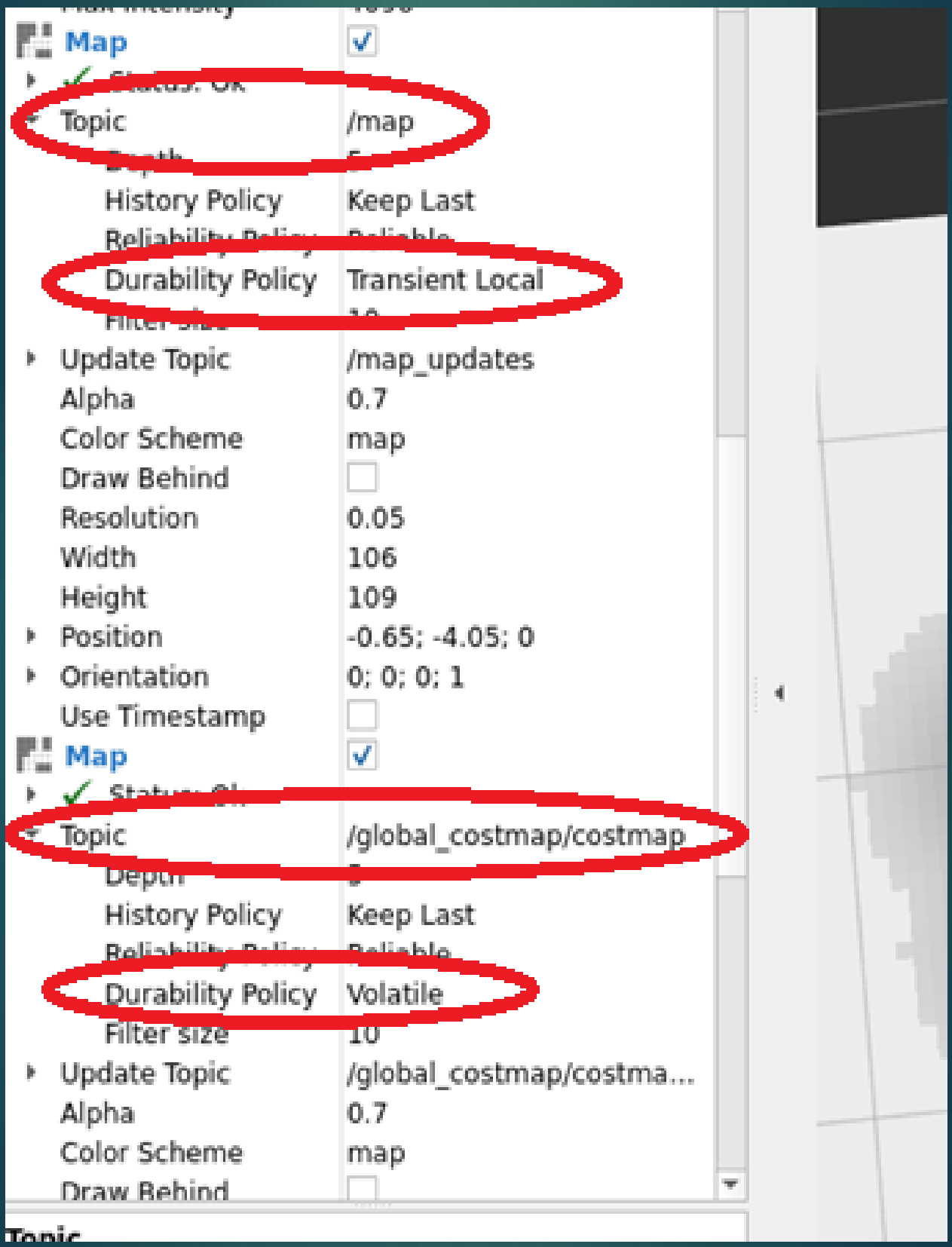
```
1. source /opt/ros/humble/setup.bash
2. ros2 launch nav2_bringup bringup_launch.py
map:=/home/user/map_name.yaml
```

- Terminal 6 (Visualization):

```
1. source /opt/ros/humble/setup.bash
2. rviz2
```

RVIZ configurations for navigation

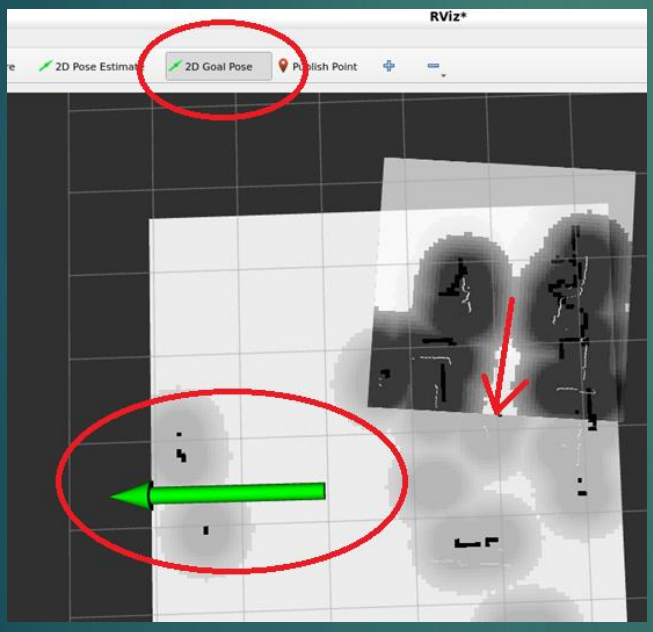
- First add Map and LaserScan and set their topics accordingly
- Then add 2 other Map displays, and set their topics to /local_costmap and /global_costmap
- Set the transient local



RVIZ configurations for navigation

- Set the 2D Pose Estimate based on where your robot is on the map in reality

- Set the 2D Goal Pose based on where you want your robot to navigat on the map



- The red arrow is the robot and direction it is looking at, and the black areas surrounding are local and global costmaps inflation layer discussed before